

Shisir Kafle

1) Soln:-

The given function is $\int_1^2 \frac{e^x}{1+x^2} \cdot dx$, $n=4$

Now,

$$a=1, b=2, n=4$$

$$\text{or } h = \frac{b-a}{n} = \frac{2-1}{4} = 0.25$$

Now,

$$x_0=1, x_1=1.25, x_2=1.5, x_3=1.75, x_4=2$$

Now,

$$f(x_0) = f(1) = \frac{e^1}{1+1^2} = \frac{e}{2} = 1.3591$$

$$f(x_1) = f(1.25) = \frac{e^{1.25}}{1+1.25^2} = 1.3620$$

$$f(x_2) = f(1.5) = \frac{e^{1.5}}{1+1.5^2} = 1.3789$$

$$f(x_3) = f(1.75) = \frac{e^{1.75}}{1+1.75^2} = 1.4165$$

$$f(x_4) = f(2) = \frac{e^2}{1+2^2} = 1.4778$$

Now,

Using Trapezoidal rule,

$$\frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$$

$$= \frac{0.25}{2} [1.3591 + 2 \times 1.3620 + 2 \times 1.3729 + 2 \times 1.4165 + 1.4778]$$

$$= \frac{0.25}{2} \times 11.1517$$

$$= \cancel{1.3939} 1.3939$$

2) Soln:-

$$f(\eta) = \int_{\pi/2}^{3\pi/2} \eta \sqrt{1 + \sin^2 \eta} \cdot d\eta, \quad h = 6$$

$$a = \frac{\pi}{2}, \quad b = \frac{3\pi}{2}, \quad h = 6$$

Now,

$$h = \frac{b-a}{6} = \frac{\frac{3\pi}{2} - \frac{\pi}{2}}{6} = \frac{2\pi}{6} = \frac{\pi}{3}$$

Now,

$$\eta_0 = \frac{\pi}{2}$$

$$\eta_5 = \frac{3\pi}{2}$$

$$\eta_1 = \pi$$

$$\eta_6 = \frac{7\pi}{2}$$

$$\eta_2 = \frac{3\pi}{2}$$

$$\eta_3 = 2\pi$$

$$\eta_4 = \frac{5\pi}{2}$$

~~Nuay~~
 ~~$f(\pi_0) = \frac{\pi}{2}$~~

Nuay,

$$\pi_0 = \frac{\pi}{2}$$

$$\pi_1 = \frac{2\pi}{3}$$

$$\pi_2 = \frac{5\pi}{6}$$

$$\pi_3 = \pi$$

$$\pi_4 = \frac{7\pi}{6}$$

$$\pi_5 = \frac{4\pi}{3}$$

$$\pi_6 = \frac{3\pi}{2}$$

Nuay,

$$f(\pi_0) = \frac{\pi}{2} \sqrt{1 + \left(\sin \frac{\pi}{2}\right)^2}$$

$$= \frac{\pi}{2} \sqrt{1 + 1}$$

$$= \frac{\pi}{2} \times \sqrt{2}$$

$$= \frac{\pi}{2} \times \sqrt{2}$$

$$= \frac{\pi}{\sqrt{2} \cdot \sqrt{2}} \times \sqrt{2} = \frac{\pi}{\sqrt{2}} = 2.2214$$

$$f(\eta_1) = \frac{2\pi}{3} \sqrt{1 + \sin^2 \eta}$$

$$= \frac{2\pi}{3} \sqrt{1 + \left(\sin\left(\frac{2\pi}{3}\right)\right)^2}$$

$$= \frac{2\pi}{3} \sqrt{1 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{2\pi}{3} \sqrt{1 + \frac{3}{4}}$$

$$= \frac{2\pi}{3} \sqrt{\frac{4+3}{4}}$$

$$= 2.7706$$

$$f(\eta_2) = \frac{5\pi}{6} \sqrt{1 + \left(\sin\left(\frac{5\pi}{6}\right)\right)^2}$$

$$= \frac{5\pi}{6} \sqrt{1 + \frac{1}{4}}$$

$$= \frac{5\pi}{6} \sqrt{\frac{4+1}{4}}$$

$$= \frac{5\pi}{6} \cdot \sqrt{\frac{5}{4}}$$

$$= 2.9270$$

$$f(\eta_3) = \pi \sqrt{1 + (\sin \pi)^2}$$

$$= \pi \sqrt{2} \quad 3.14159$$

$$f(\eta_4) = \frac{7\pi}{6} \sqrt{1 + \left(\sin \frac{7\pi}{6}\right)^2}$$

$$= 4.097$$

$$f(\eta_5) = \frac{4\pi}{3} \sqrt{1 + \left(\sin \frac{4\pi}{3}\right)^2}$$

$$= 5.541$$

$$f(\eta_6) = \frac{3\pi}{2} \sqrt{1 + \left(\sin \left(\frac{3\pi}{2}\right)\right)^2}$$

$$= 6.664$$

Now
Using Simpson's $1/3$ rule.

$$\frac{h}{3} [f(\eta_0) + 4f(\eta_1) + 2f(\eta_2) + 4f(\eta_3) + 2f(\eta_4) + 4f(\eta_5) + f(\eta_6)]$$

$$\frac{h}{3} [2.2214 + 4 \times 2.7706 + 2 \times 2.9270 + 4 \times 3.14159 + 2 \times 4.097 + 4 \times 5.541 + 6.664]$$

$$= \frac{h}{3} [68.74620] \times \frac{\pi}{6 \times 3}$$

$$= \frac{\pi}{18} \times 68.74620$$

$$= 11.99$$

3) Soln:-

$$y' = x + y + xy, \quad y(0) = 1$$

Now,
 $x_0 = 0, y_0 = 1$

Using Euler method (Iteration)

Iteration 1
 $y_{i+1} = y_i + f(x_i, y_i) \times h$

$$y_{0+1} = y_0 + f(x_0, y_0) \times h$$

$$y_1 = 1 + (0 + 1 + 1 \times 0) \times 0.5$$

$$y_1 = 1 + (1) \times 0.5$$

$$y_1 = 1.5, \quad x_1 = 0.5$$

Iteration 2

$$y_2 = y_1 + f(x_1, y_1) \times h$$

$$y_2 = 1.5 + (0.5 + 1.5) \times 0.5$$

$$y_2 = 1.5 + (0.5 + 1.5 + 0.5 \times 1.5) \times 0.5$$

$$y_2 = 2.875, \quad x_2 = 1$$

Iteration 3

$$y_3 = y_2 + f(x_2, y_2) \times h$$

$$= 2.875 + (1 + 2.875) \times 0.5$$

$$= 2.875 + (1 + 2.875 + 1 \times 2.875) \times 0.5$$

$$= 6.25, \quad x_3 = 1.5$$

Iteration 4

$$y_4 = y_3 + f(x_3, y_3) \times h$$

$$y_4 = 6.25 + f(1.5, 6.25) \times 0.5$$

$$y_4 = 6.25 + (1.5 + 6.25 + 6.25 \times 1.5) \times 0.5$$

$$y_4 = 14.8125, \quad x_4 = 2$$

$\therefore$ When x_4 is 2 then the value of $y(2)$ is 14.8125

$$\therefore y(2) = 14.8125$$

4) Soln:-

The given function $f(x) = 3 \sin x - x^2$

Now
 $x_0 = 3, \quad x_L = -2$

Now
Find x_1 & x_2

$$d = \frac{\sqrt{5}-1}{2} (3 - (-2))$$

$$d = \frac{\sqrt{5}-1}{2} \times 5$$

$$d = 3.090$$

$$x_1 = x_0 + d = -2 + 3.090 = 1.09$$

$$x_2 = x_0 - d = 3 - 3.090 = -0.09$$

Now

left hand ko hula

$$f(\eta_1) = 3 \sin(1.09) - \frac{(1.09)^2}{5}$$

$$= 2.4222$$

$$f(\eta_2) = 3 \sin(-0.09)$$

$$f(\eta) = 3 \sin \eta - \frac{\eta^2}{5}$$

$$3 \cos \eta - \frac{2\eta}{5}$$

$$\eta_{n+1} = \eta_n - \frac{f'(\eta)}{f''(\eta)}$$

$N_{4,}$

$$\eta \in [0, 3]$$

$N_{4,}$

$$\eta_0 = \frac{0+3}{2} = \frac{3}{2} = 1.5$$

$N_{4,}$

$$\eta_0 = \eta_0 - \frac{3 \sin(1.5) - \frac{(1.5)^2}{5}}{3 \cos \eta - \frac{2\eta}{5}}$$

$$3 \cos \eta - \frac{2\eta}{5}$$

$$\eta_1 = 1.5 - \frac{3 \sin(1.5) - \frac{(1.5)^2}{5}}{3 \cos \eta - \frac{2\eta}{5}}$$

4) Gilph

$$f(\eta) = 3 \sin(\eta) - \frac{\eta^2}{5}$$

$$\eta_2 = 0 \text{ \& } \eta_4 = 3$$

Interval

$$d = \frac{\sqrt{5}-1}{2} (3-0) = 1.854$$

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4) Soln:-

$$\eta_1 = \eta_0 + d = 0 + 1.854 = 1.854$$

$$\eta_2 = \eta_0 - d = 3 - 1.854 = 1.146$$

$$F(\eta_1) = F(1.854) = 3 \ln(1.854) - \frac{(1.854)^2}{5}$$

$$F(\eta_2) = F(1.146) = 3 \ln(1.146) - \frac{(1.146)^2}{5}$$

$$= 2.470$$

Since $F(\eta_2) > F(\eta_1)$

$$\eta_4 = 1.854$$

Iteration 2

$$\eta_L = 0, \eta_U = 1.854$$

$$d = \frac{\sqrt{5}-1}{2} (1.854 - 0) = 1.145$$

$$\eta_1 = 0 + 1.145 = 1.145$$

$$\eta_2 = 1.854 - 1.145 = 0.709$$

$$F(1.145) = 3 \ln(1.145) - \frac{(1.145)^2}{5} = 2.465$$

$$F(0.709) = 3 \ln(0.709) - \frac{(0.709)^2}{5} = 1.852$$

Since $F(\eta_1) > F(\eta_2)$

$$\eta_L = 0.709$$

Iteration 3

$$\eta_1 = 0.709, \eta_U = 1.854$$

$$d = \frac{\sqrt{5}-1}{2} (1.854 - 0.709) = 0.707$$

$$\eta_1 = 0.709 + 0.707 = 1.416$$

$$\eta_2 = 1.854 - 0.707 = 1.147$$

$$f(1.416) = 3 \sin(1.416) - \frac{(1.416)^2}{5} = 2.563$$

$$f(1.147) = 3 \sin(1.147) - \frac{(1.147)^2}{5} = 2.471$$

Since,

$$f(\eta_1) > f(\eta_2)$$

$$\eta_1 = 1.147$$

Step 4:

$$\eta_3 = 1.147, \eta_4 = 1.854$$

$$d = \frac{\sqrt{5}-1}{2} (1.854 - 1.147) = 0.436$$

$$\eta_1 = 1.147 + 0.436 = 1.583$$

$$\eta_2 = 1.854 - 0.436 = 1.418$$

$$f(1.583) = 3 \sin(1.583) - \frac{(1.583)^2}{5} = 2.498$$

$$f(1.418) = 3 \sin(1.418) - \frac{(1.418)^2}{5} = 2.562$$

Since $f(\eta_2) > f(\eta_1)$

$$\eta_4 = 1.583$$

$$\therefore \eta_3 = 1.147 \text{ \& } \eta_4 = 1.583$$

Now

$$\eta_{max} = 1.365$$

$$= 3 \sin(1.365) - \frac{(1.365)^2}{5} = 2.564$$

$\therefore$ Maximum value is 2.564 at $\eta = 1.365$